

Beyond the Gatekeeper: A Framework for AI-Assisted Open Peer Review and Decentralized Scientific Evaluation

A position and systems-design preprint

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Abstract. Scientific peer review performs indispensable functions, but it is also a scarce human-attention mechanism operating under increasing submission volume, heterogeneous research communities, uneven access to publishing infrastructure, and persistent concerns about cost, delay, reviewer burden, bias, and reliability. At the same time, large language models (LLMs) and other AI systems have become capable of producing structured critiques, performing literature retrieval, generating code, executing computational checks, and coordinating multi-agent evaluation workflows. This paper proposes *AI-assisted open peer review*: a deliberately non-replacement architecture in which open preprints remain freely disseminable, a heterogeneous panel of AI evaluators performs blind and evidence-oriented preliminary scrutiny, tool-assisted agents attempt verification and falsification, disagreements are surfaced rather than averaged away, and promising or inconclusive manuscripts are routed to human experts for deeper evaluation. The proposal further develops a decentralized publication model in which journal publication becomes one possible certification layer rather than the sole gateway through which a research object can acquire a persistent record of scrutiny. The central claim is not that AI can determine scientific truth, replace expert mathematicians, or serve as an autonomous acceptance oracle. Rather, AI can plausibly serve as a scalable *triage and evidence-assembly layer* that reduces the amount of routine screening demanded from scarce human experts while preserving human authority over consequential judgments. The paper formalizes this architecture, proposes multi-model disagreement and adversarial review protocols, introduces a persistent critique ledger and evidence record, specifies safeguards against correlated model error, hallucination, prompt injection, prestige bias, confidentiality failures, and reviewer gaming, and defines an empirical benchmark for testing whether such a system improves recall of valuable work without introducing a new computational gatekeeper. Particular attention is paid to independent and early-career researchers whose access to conventional publication infrastructure may be constrained by affiliation, geography, cost, or professional status. The resulting framework treats scientific evaluation as a distributed evidence process rather than a binary publication event.

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1 Introduction

Scientific knowledge advances through a mixture of discovery, criticism, replication, communication, and accumulation. Peer review occupies an important position in this system because it asks knowledgeable humans to scrutinize research before it is formally certified by a venue. Yet the conventional model is under visible strain. Submission volumes have grown rapidly in several research communities, reviewer labor is finite, and a review is itself an expensive intellectual activity. The result is a structural tension: the scientific community would like increasingly thorough evaluation at precisely the time when the supply of qualified human attention is least elastic.

This tension creates a second problem that is less frequently expressed in computational terms. A research manuscript may be intellectually valuable without immediately being valuable *to a particular journal*. The predicates

“deserves serious attention” and “should be accepted by venue V ”

are not logically identical. Journals perform several functions at once: dissemination, curation, evaluation, certification, archival organization, and community signaling. Bundling these functions is historically understandable, but it also means that a researcher whose work is difficult to place, whose field is interdisciplinary, or who lacks institutional access may encounter barriers unrelated to the underlying intellectual quality of the manuscript.

The present paper proposes a modest but consequential intervention: introduce an additional evaluation layer between open dissemination and conventional expert peer review. The layer is called *AI Review*. The name is intentionally simple, but the proposed system is not a simplistic chatbot that answers “good” or “bad.” It is a structured, multi-stage, heterogeneous evaluation pipeline intended to produce an auditable dossier of strengths, objections, uncertainty, literature context, and verification attempts.

The proposal is motivated by three observations.

First, AI-assisted review is no longer purely hypothetical. AAAI-26 reported a field deployment in which every main-track submission receiving full review was additionally processed by an AI review system, covering 22,977 papers in less than a day; the system combined frontier models, tool use, and safeguards rather than relying on a single prompt [1]. This does not establish that AI can replace experts, but it does establish that large-scale machine-assisted evaluation is operationally plausible.

Second, heterogeneous panels matter. A 2026 study in *Scientometrics* evaluated panels containing GPT-4.1, Claude Sonnet 4, Gemini 2.5 Flash, Kimi K2.5, Llama 3.3 70B, and Qwen 3.5 397B. The study found that comparative selection across a heterogeneous panel could produce more stable alignment with known venue outcomes than simple score thresholds applied to individual models [2]. This result directly motivates the present paper’s emphasis on *model diversity, comparative judgment, and disagreement* rather than one universal model score.

Third, the limitations of AI reviewing are real. Recent work has documented institutional-affiliation bias in LLM-assisted review: changing author metadata while holding manuscript content fixed can alter judgments in the direction of prestige [3]. Other work has argued that AI review should move beyond fluent review generation toward reproducibility checking, citation verification, and other tool-assisted tasks [5, 6]. LLMs also continue to produce confident falsehoods, making unsupported machine judgment an unsafe foundation for scientific certification [7].

These observations lead to the central thesis of this paper:

Design Principle 1.1. *AI should not be proposed as a replacement for scientific peer review. It should be proposed as a scalable, heterogeneous, blind, tool-assisted triage and evidence layer that helps allocate scarce human expert attention, while preserving human review, author response, appeal, and optional formal journal certification.*

The distinction is crucial. The system proposed here is not designed to answer the meta-physical question “Is this true?” It is designed to answer a narrower operational question: *given limited human attention, how can a research community identify manuscripts that warrant serious expert scrutiny, identify obvious weaknesses before that scrutiny occurs, and make the resulting evaluation process more transparent and less dependent on prestige signals?*

The remainder of the paper develops this idea as a systems architecture rather than an assertion of technological inevitability.

2 Scope, Terminology, and Normative Position

2.1 What this paper is and is not

This manuscript is a position and systems-design paper. It does not claim to have demonstrated that a particular AI review system is accurate enough to certify research. It does not report a new controlled benchmark of the author’s own implementation. It does not claim that present-day LLMs understand mathematics, science, or scholarly novelty in the same sense as domain experts. Its contribution is architectural: define a concrete model that can be implemented and falsified, identify the principal risks, and specify measurable criteria by which the proposal should succeed or fail.

The paper uses “AI” primarily to refer to modern computational evaluation systems that may include LLMs, retrieval systems, symbolic engines, numerical software, proof assistants, code execution environments, and other specialized tools. An LLM is therefore only one component of the proposed evaluator.

2.2 AI Review versus peer review

The term *peer review* has institutional and sociological meanings as well as a functional meaning. To avoid falsely equating a machine output with expert judgment, this paper uses the term *AI Review* for the preliminary machine-assisted layer and reserves *expert review* for consequential human evaluation.

Accordingly,

$$\text{AI Review} \neq \text{Human Peer Review}.$$

The relationship proposed here is additive:

$$\text{Open Preprint} \longrightarrow \text{AI Review} \longrightarrow \text{Expert Review} \longrightarrow \text{Optional Certification}.$$

2.3 The two separations

The proposal depends on two conceptual separations that are often conflated.

Dissemination versus evaluation. A manuscript can be disseminated without first receiving institutional approval. A preprint server already provides one mechanism for this. AI Review adds a machine-assisted evaluation record but does not need to control whether the manuscript exists.

Evaluation versus certification. An expert may judge a theorem correct, useful, and novel without that judgment becoming a publication certificate. Conversely, journal acceptance may reflect venue fit and editorial considerations in addition to scientific quality. The proposed architecture therefore treats certification as one possible endpoint rather than the unique gateway to serious attention.

3 The Scientific-Workflow Problem as an Allocation Problem

A useful abstraction is to treat scientific review as an allocation problem under finite expert attention.

Let a research community receive manuscripts

$$\mathcal{P} = \{P_1, P_2, \dots, P_N\}.$$

Let the available expert-review capacity be

$$B_H > 0,$$

measured, for example, in reviewer-hours per unit time. If each manuscript requires expected human effort h_i , then the community can deeply evaluate only a subset satisfying roughly

$$\sum_{P_i \in S} h_i \leq B_H.$$

The optimization problem is not merely to maximize the number of reviewed papers. A better objective is to maximize the expected scientific value uncovered by scarce human attention:

$$\max_{S \subseteq \mathcal{P}} \sum_{P_i \in S} U(P_i), \quad \text{subject to} \quad \sum_{P_i \in S} h_i \leq B_H,$$

where $U(P_i)$ is an unobserved quantity representing the potential value of expert scrutiny for manuscript P_i .

An AI system cannot be expected to know $U(P_i)$ exactly. The practical objective is therefore to produce an inexpensive proxy $\hat{U}(P_i)$ that is useful for routing human attention.

This reframing clarifies why a triage architecture can be valuable even if AI is imperfect. If a machine can eliminate obvious defects, retrieve relevant literature, reproduce computational claims, or prioritize manuscripts with credible evidence of novelty at low marginal cost, then the system may improve the efficiency of human review without replacing its epistemic role.

The relevant performance criterion is therefore not:

$$\mathbb{P}(\text{AI decision} = \text{ultimate truth}),$$

but rather a vector of quantities such as:

(human-hours saved, valuable-work recall, false-negative rate, verification accuracy, bias, cost).

This is a central design choice. The paper's proposed system is not an oracle; it is a resource-allocation mechanism.

4 Formal Definition of AI-Assisted Open Peer Review

Definition 4.1 (Research object). *A research object P is a versioned manuscript together with any declared supplementary material required to assess its claims, such as data, source code, proofs, appendices, computational notebooks, or experimental protocols.*

Definition 4.2 (AI Review). *AI Review is a structured computational process that receives a research object P , evaluates it under a predefined rubric using one or more heterogeneous AI systems and verification tools, records model-level judgments and disagreements, and produces an auditable evaluation report for downstream human or community scrutiny.*

A useful representation is

$$\text{AIReview}(P) = (Q(P), K(P), V(P), D(P), U(P), L(P)),$$

where:

- $Q(P)$ is a multidimensional quality profile;
- $K(P)$ is a ledger of identified criticisms;
- $V(P)$ records attempted verification;
- $D(P)$ summarizes disagreement among evaluators;
- $U(P)$ records uncertainty or abstention;
- $L(P)$ records literature and provenance checks.

4.1 A multidimensional quality vector

Let

$$Q(P) = (C, N, S, G, E, R),$$

where, depending on discipline,

- C = correctness,
- N = novelty or distinctiveness,
- S = significance,
- G = rigor,
- E = evidentiary support,
- R = reproducibility or auditability.

The entries should be accompanied by confidence and evidence rather than treated as exact measurements. A mathematical paper may make C and proof verification dominant. A clinical paper may make methodology, statistical validity, and ethical compliance dominant. A computational paper may additionally require executable artifact verification.

Design Principle 4.1 (No universal scalar score). *A serious AI Review system should not reduce a multidimensional scientific evaluation to one unqualified number. Scalar summaries may be presented for convenience, but the underlying profile, critical objections, evidence, disagreement, and uncertainty must remain visible.*

5 Why Heterogeneous Models Rather Than a Single AI?

Suppose $M_1, \dots, M_k$ are evaluation systems. A naïve design computes

$$\bar{S}(P) = \frac{1}{k} \sum_{j=1}^k S_j(P)$$

and accepts manuscripts above a threshold. The problem is that numerical scores are not necessarily calibrated across models. One model may use a narrow range; another may be generous; a third may be unusually severe.

More importantly, model errors may be correlated. If all systems have learned similar conventions from similar literature, then agreement may reflect shared blind spots rather than independent evidence. Thus,

model diversity $\neq$ statistical independence.

The appropriate role of heterogeneity is therefore not to create a fake “jury of six humans” but to increase the diversity of hypotheses about the manuscript.

Let each evaluator produce an evidence-bearing report

$$R_j(P) = (Q_j, K_j, V_j, U_j).$$

The panel then computes not only a consensus profile but also disagreement statistics such as

$$\Delta_Q(P) = \text{Dispersion}(Q_1(P), \dots, Q_k(P)).$$

A high $\Delta_Q(P)$ should itself be an output. In particular, the system should never conceal disagreement by arithmetic averaging alone.

Recent empirical work is especially relevant here. Douven’s 2026 study compared heterogeneous cloud and open-weight LLM panels and found that comparative selection yielded 78% decision alignment for the cloud panel and 74% for the open-weight panel on a set of ICLR papers, while fixed score thresholds depended strongly on which model happened to be best calibrated [2]. The appropriate inference is not that “LLMs can replace reviewers,” but that *panel structure and comparative evaluation can matter substantially*.

5.1 Model portfolio design

A practical panel may include several general frontier models, several open-weight models, and specialized verification agents. The exact model roster should change over time; the architecture should not depend on any one vendor.

A desirable panel contains diversity along several dimensions:

1. training and alignment lineage;
2. model architecture and scale;
3. proprietary versus open-weight access;
4. tool-use capability;
5. symbolic or numerical specialization;

6. prompt and reasoning strategy.

The system should record exact model versions because model behavior changes over time. A review made in 2026 should remain distinguishable from a review made by a later model in 2028.

6 Role-Based AI Referees

Rather than assigning every model the same vague request, AI Review should decompose evaluation into specialized roles.

6.1 The correctness referee

The correctness referee attempts to identify logical, mathematical, statistical, or methodological errors. In mathematics, it should inspect quantifiers, hypotheses, boundary cases, definitions, proof dependencies, branch conditions, limiting operations, and hidden assumptions.

6.2 The novelty referee

The novelty referee searches the literature and constructs nearest-neighbor comparisons. It should not merely say “this seems new.” It should report:

Claim $\rightarrow$ Nearest prior work $\rightarrow$ Difference $\rightarrow$ Materiality of difference.

A novelty judgment should therefore be a comparison task, not a memory task.

6.3 The adversarial referee

This referee is explicitly instructed to try to destroy the manuscript. It seeks counterexamples, false generalizations, missing hypotheses, incompatible assumptions, and failure of stated algorithms.

6.4 The significance referee

This referee asks whether the central result changes understanding, enables an application, settles a question, improves an algorithm, or offers a useful conceptual framework.

6.5 The reproducibility referee

This referee attempts to execute code, reconstruct calculations, inspect data, reproduce numerical tables, or identify missing information.

6.6 The exposition referee

This referee checks whether the paper actually states what it means, distinguishes theorem from conjecture, identifies assumptions, and avoids ambiguity.

The role decomposition is important because a general-purpose LLM is often too eager to produce a smooth overall judgment. Specialization increases the chance that the system finds concrete, falsifiable objections.

7 Adversarial Cross-Examination

A central feature of the proposed architecture is an adversarial exchange among evaluators.

Let the initial panel generate criticism sets

$$K_1, K_2, \dots, K_k.$$

For each criticism $c \in K_j$, another evaluator must independently assess it. The system then creates a local dispute graph:

$$c \longrightarrow \text{evidence} \longrightarrow \text{response} \longrightarrow \text{adjudication}.$$

The objective is not to force consensus. It is to determine whether a disagreement corresponds to a real unresolved scientific issue.

For example, if one evaluator states:

Theorem 3 is false.

the system should not record “negative review.” It should request a structured objection:

1. identify the exact claim;
2. identify the exact location in the manuscript;
3. state the logical failure;
4. provide a counterexample, calculation, prior theorem, or other evidence;
5. state whether the issue is fatal, major, or minor.

A second agent then attempts to validate the objection. A third agent may adjudicate if disagreement persists.

Design Principle 7.1 (Argument over impression). *An AI critique should be evaluated primarily as an argument supported by evidence, not as a prose artifact. Polished language, confidence, harshness, or verbosity must not by themselves determine the weight of a criticism.*

8 Tool-Assisted Verification

The proposal becomes substantially stronger when language models are used as orchestration layers around verifiers rather than as sole sources of truth.

8.1 Mathematics

For mathematical manuscripts, verification may include symbolic algebra, numerical exploration, automated counterexample search, and, where feasible, formal proof assistants. A model might generate a conjectured identity, but a computer algebra system can test it over symbolic domains or random instances; a proof assistant may verify a formalized lemma.

The workflow is therefore:

$$\text{LLM hypothesis} \rightarrow \text{formal/symbolic test} \rightarrow \text{human-readable evidence}.$$

8.2 Code and computational science

A computational claim should be executable where possible. The evaluator can:

read code → install isolated dependencies → execute → compare outputs → record discrepancies.

Sandboxed execution is essential: manuscript code must be treated as untrusted input.

8.3 Statistics

A statistical referee can independently recompute descriptive statistics, confidence intervals, likelihoods, effect sizes, or regression outputs from the supplied data. The LLM then explains the discrepancy if the numbers disagree.

8.4 Literature verification

Citations should be checked against primary sources where feasible. The system should verify that a cited work exists, that the title and bibliographic metadata are correct, and that the cited source actually supports the claim attributed to it.

An important 2026 position paper argues for exactly this shift from mere review-text generation toward tasks such as reproducibility verification and citation correctness checking [6]. The present proposal generalizes that insight into a complete evaluation architecture.

9 The Three-Valued Triage Principle

The proposed system should resist binary machine gatekeeping. We define three primary outcomes:

Definition 9.1 (Triage statuses). *For manuscript P , an AI Review system may assign:*

1. **Promising:** *evidence is sufficiently favorable that expert attention is recommended;*
2. **Problematic:** *substantial defects were identified and documented;*
3. **Inconclusive:** *the system lacks adequate confidence or competence to classify the work reliably.*

The third state is fundamental. An AI system that does not understand a specialist theorem should not convert uncertainty into rejection.

A safer decision rule is therefore:

$$AI(P) = \begin{cases} \text{escalate,} & \text{promising,} \\ \text{escalate with warnings,} & \text{inconclusive,} \\ \text{revise or challenge,} & \text{problematic.} \end{cases}$$

The system should generally optimize for high recall of potentially valuable work. A false positive consumes some human attention. A false negative can prevent a potentially valuable result from receiving any expert scrutiny.

This asymmetry is especially important for unconventional research.

10 Blind and Prestige-Neutral Evaluation

A distinctive feature of the proposed architecture is that the first AI pass should be blind by default.

The initial evaluator should receive the manuscript and necessary technical materials, but should not receive, where operationally feasible:

- author name;
- institution and laboratory;
- academic degree or career stage;
- nationality or location;
- citation count;
- awards or social status;
- other prestige-signaling metadata.

This is not because identity is never relevant. Conflict-of-interest checks, authorship verification, ethics, or provenance may legitimately require identity later. The principle is simply temporal: *content should be evaluated before prestige information is introduced*.

The need for this safeguard is supported by Vasu et al., who experimentally varied author metadata while holding paper content fixed and found systematic affiliation effects in LLM-assisted review, including stronger judgments for papers associated with highly ranked institutions [3].

10.1 Counterfactual bias audit

A deployed system should periodically run controlled counterfactuals:

$$P + \text{Metadata}_A \quad \text{versus} \quad P + \text{Metadata}_B,$$

where the manuscript is identical and only the author profile changes.

Define the metadata sensitivity

$$B_{\text{meta}} = \mathbb{E} [|S(P, m_A) - S(P, m_B)|].$$

A useful evaluation system should drive B_{meta} toward zero for irrelevant metadata while preserving legitimate sensitivity to information that genuinely affects scientific validity, such as disclosed conflicts of interest or missing authorship information when those are directly relevant.

11 Handling Hostile, Elitist, or Status-Based Reviews

Human peer review is a form of argumentation. Reviewer tone and reviewer validity should therefore be modeled separately.

Suppose a reviewer writes:

The author’s central theorem is false because equation (17) assumes positivity, whereas the theorem quantifies over all real inputs.

This may be technically valuable even if the reviewer is rude.

Conversely:

The author has no PhD and therefore should not make claims in this area.

may be a status-based judgment rather than a scientific objection.

The system should therefore decompose reviewer feedback into at least two independent dimensions:

technical validity and identity/status relevance.

A comment can simultaneously be:

technically valid + improperly framed,

or

technically invalid + polite.

The AI should not suppress negative reviews merely because they are hostile. It should determine whether the reviewer has supplied a substantively checkable objection.

Design Principle 11.1 (No credential veto). *The absence of a conventional credential may not itself constitute evidence that a mathematical, scientific, or technical claim is false. Credentials can be relevant to biography, expertise, conflict-of-interest analysis, or institutional certification, but they should not substitute for an argument about the research object.*

This principle is especially important for open research environments. It does not claim that outsiders are superior to insiders. It claims only that scientific criticism should be attached to the content being criticized.

12 The Critique Ledger

We propose that every reviewed research object maintain a persistent *critique ledger*.

A minimal schema is:

Claim	Objection	Evidence	Resolution
Theorem 2	Missing hypothesis	Counterexample	Manuscript revised
Novelty claim	Prior work found	Bibliographic comparison	Claim narrowed
Numerical result	Reproduction mismatch	Independent execution	Code corrected
Definition 4	Ambiguous domain	Formal check	Definition rewritten

The ledger should be versioned rather than overwritten. This creates a persistent history of scientific disagreement and resolution.

The status of a manuscript can therefore evolve from

$$P^{(1)} \rightarrow P^{(2)} \rightarrow P^{(3)} \rightarrow \dots$$

with corresponding review records

$$R^{(1)}, R^{(2)}, R^{(3)}, \dots$$

This transforms peer review from a hidden event into a cumulative evidence process.

13 From AI Review to Human Expert Review

The proposed architecture does not eliminate experts; it changes how they are allocated.

Let each manuscript receive an AI triage report $A(P)$. An expert-routing function

$$\rho : A(P) \mapsto \mathcal{E}$$

selects an appropriate class of human expertise.

Examples include:

- a number theorist for an arithmetic conjecture;
- a formal methods expert for a proof-assistant result;
- a statistician for a high-dimensional inference claim;
- a control engineer for an engineering model;
- a domain scientist for a specialized experimental result.

The AI system should recommend expertise, not manufacture expertise.

13.1 Positive expert signals

A human expert may provide one of several evidence types:

1. confirmation of a specific theorem or calculation;
2. identification and resolution of a specific flaw;
3. confirmation that the novelty claim survives literature comparison;
4. successful independent reproduction;
5. substantive conceptual endorsement with explicit reasoning.

A bare “like” or anonymous vote is weaker evidence than a technically detailed assessment. The platform should therefore distinguish social endorsement from expert verification.

13.2 The right to human appeal

A defining safeguard is that authors should be able to request human attention when the system is uncertain or when a machine objection is disputed.

Design Principle 13.1 (Right to human appeal). *AI Review may prioritize, challenge, or recommend revision, but it should not possess an irreversible veto over access to human scientific scrutiny.*

This rule prevents the proposed anti-gatekeeping mechanism from becoming a new gatekeeper.

14 Decentralized Scientific Evaluation

The deeper institutional proposal is to decouple scientific evaluation from any single publication venue.

The conventional workflow can be represented abstractly as

Author $\rightarrow$ Journal $\rightarrow$ Editor $\rightarrow$ Reviewers $\rightarrow$ Decision.

The proposed workflow is

Author $\rightarrow$ Open Preprint $\rightarrow$ Blind AI Review $\rightarrow$ Expert Review $\rightarrow$ Evidence Record $\rightarrow$ Optional Journal Certification

The advantage is not that journals become useless. Rather, the journal becomes one possible certification layer among several mechanisms for establishing trust.

A research object could carry a persistent state such as:

Research state	Example evidence
Open	Preprint publicly available
AI-screened	Heterogeneous AI report archived
Technically challenged	Specific objections logged
Expert-reviewed	Named or verified expert review available
Reproduced	Independent computational or experimental reproduction
Certified	Journal or society certification

These states are not mutually exclusive. A paper may be expert-reviewed without being journal-published, or journal-published while continuing to accumulate post-publication evidence.

This architecture is consistent with the broader movement toward open peer review and decentralized scholarly infrastructure, including platforms such as OpenReview that support public review and discussion and research literature exploring decentralized science systems [9, 10].

15 Why This May Help Independent Researchers

The proposal has a particular relevance to researchers who lack conventional institutional infrastructure.

An independent researcher may encounter barriers involving:

- access to subscription literature;
- publication charges;
- lack of a laboratory or research group;
- lack of senior colleagues who can provide informal pre-review;
- difficulty finding referees willing to read unconventional work;
- geographic and institutional visibility;
- signaling effects associated with unfamiliar affiliations.

These barriers do not imply that the conventional academic system is scientifically illegitimate. They imply that institutional publication status is an imperfect proxy for the quality of an individual research object.

The proposal therefore adopts a neutral standard:

No discount for outsiders. No penalty for outsiders.

A paper should succeed or fail on its scientific arguments. The infrastructure should make that principle easier to approximate.

This section must be read as a design motivation, not as evidence that outsiders are inherently superior. Historical examples of important work produced outside conventional academic pathways demonstrate that institutional credentials and intellectual capability are not identical variables, but they do not establish a universal “outsider advantage.” The present architecture therefore targets *status-insensitive access to scrutiny*, not preferential treatment for independent researchers.

16 Traditional Peer Review as a Certification Layer

A major source of resistance to decentralized evaluation is the fear that removing journals removes standards. This paper rejects that inference.

Traditional peer review can remain valuable for:

- disciplinary curation;
- editorial coherence;
- formal certification;
- archival organization;
- conflict-of-interest management;
- community-specific quality thresholds;
- prestige and signaling, where legitimately earned.

The proposal is therefore evolutionary rather than abolitionist:

Current system + AI triage + open evidence records = richer scientific infrastructure.

A journal could even choose to accept external AI Review reports as additional editorial evidence. Another journal could ignore them. A society could build its own secure review network. A university could operate a domain-specific expert pool.

The scientific ecosystem becomes modular.

17 Why Raw AI Scores Are Insufficient

A tempting implementation is:

$$S(P) = \frac{1}{k} \sum_{j=1}^k S_j(P),$$

with an acceptance threshold $S(P) \geq T$.

This design should not be the default.

First, scores may be poorly calibrated across models. Second, a single fatal mathematical flaw can coexist with excellent style and significance scores. Third, high disagreement is hidden by averaging. Fourth, the numerical range of scores creates false precision.

We therefore propose a structured decision function:

$$D(P) = F(Q(P), K(P), V(P), \Delta(P), U(P)),$$

where F is a routing rule rather than an acceptance oracle.

A manuscript with scores

$$(90, 91, 88, 89, 92)$$

should not automatically pass if one evaluator has found a credible counterexample to the central theorem. Conversely, a manuscript with scores

$$(52, 55, 49, 53, 57)$$

should not automatically be discarded if all evaluators agree that they lack expertise to assess its specialized mathematics.

The correct unit of analysis is therefore not the score but the *evidence bundle*.

18 Novelty as a Retrieval and Comparison Problem

Novelty is often the weakest part of automated scientific evaluation because a model may recognize familiar language without establishing whether the exact contribution has already appeared.

A robust novelty module should construct an evidence graph:

$$P \rightarrow \{C_1, \dots, C_m\} \rightarrow \{L_1, \dots, L_n\} \rightarrow \text{comparison matrix}.$$

Here C_i are the manuscript's major claims and L_j are relevant prior works.

For each important claim, the system should report:

1. nearest prior result;
2. exact overlap;
3. exact difference;
4. whether the difference changes the theorem, method, application, or interpretation;
5. whether the author's novelty language should be narrowed.

The output might say:

The general identity is known. The manuscript's novelty, if any, lies in the restricted domain, computational consequence, or new theorem stated in Section 4.

This form of criticism is valuable even when the result is ultimately judged non-novel. It converts an embarrassing discovery during journal review into a useful correction before submission.

19 Threat Model

A credible AI Review system needs an explicit threat model. The following failure modes are considered first-class risks.

19.1 Correlated model error

Multiple models may share the same misconception. Panel agreement is not proof.

19.2 Hallucinated evidence

LLMs can produce plausible but false statements and fabricated citations. The 2026 *Nature* study by Kalai et al. emphasizes that confident falsehoods remain a fundamental reliability problem [7].

19.3 Prompt injection

A manuscript may contain text that tries to manipulate its evaluator. Review systems must treat manuscripts as untrusted input, isolate instructions from manuscript content, and use sandboxed tools. Recent literature on AI review security explicitly identifies prompt injection and manipulation as relevant concerns [8].

19.4 Prestige bias

Models can favor prestigious affiliations or senior authors. Counterfactual metadata audits are therefore required [3].

19.5 Novelty blindness

Highly unconventional work may resemble nonsense to an evaluator trained on prevailing conventions. The inconclusive state and human appeal channel are designed to mitigate this.

19.6 Goodhart effects

Researchers may optimize manuscripts for the evaluation system. Once the rubric becomes a target, superficial compliance can replace genuine quality.

19.7 Model monoculture

If an entire field uses the same evaluator, errors become system-wide. Heterogeneous models and open benchmarking are therefore governance requirements, not merely technical conveniences.

19.8 Confidentiality

Unpublished manuscripts may contain sensitive material. Reviewers should not upload confidential manuscripts to arbitrary third-party models. Systems must provide secure inference pathways, local or controlled models where appropriate, explicit data-retention policies, and author consent for machine processing.

19.9 Automation bias

Humans may accept AI criticism simply because it looks technical. The interface should require evidence and disclose uncertainty rather than presenting machine judgments as facts.

20 Security and Governance Architecture

A secure deployment should follow a trust hierarchy.

$$\boxed{\text{Manuscript}} \rightarrow \boxed{\text{Sandbox}} \rightarrow \boxed{\text{Evaluator}} \rightarrow \boxed{\text{Verifier}} \rightarrow \boxed{\text{Human}}.$$

No manuscript should be allowed to modify the system prompt, access arbitrary external credentials, execute uncontrolled code, or exfiltrate reviewer information.

Every AI Review record should contain:

- model family and exact version;
- evaluation date;
- rubric version;
- tools used;
- literature sources accessed;
- verification attempts;
- model-level outputs;
- adjudication outputs;
- uncertainty and abstention information.

A future-proof system should permit the original review to be replayed or audited when practical.

21 A Formal Consensus-and-Disagreement Record

Let each evaluator j return:

$$R_j = (Q_j, K_j, V_j, U_j).$$

Define a consensus estimator for a quality dimension d by

$$\tilde{Q}_d = \text{median}_{j=1}^k Q_{j,d},$$

which is often more robust to extreme reviewers than a mean. Let the disagreement be

$$\Delta_d = \text{MAD}_{j=1}^k(Q_{j,d}),$$

where MAD denotes median absolute deviation.

But even robust aggregation must not override a single strong falsification. We therefore define a critical-objection flag

$$F(P) = 1$$

if at least one objection satisfies a predetermined evidence threshold, such as a verified counterexample or a reproducibly demonstrated methodological failure.

A simple routing policy is then:

$$\text{Route}(P) = \begin{cases} \text{Human verification required,} & F(P) = 1, \\ \text{Expert attention recommended,} & F(P) = 0, \tilde{Q} \text{ favorable,} \\ \text{Human attention recommended,} & \Delta \text{ high or } U \text{ high,} \\ \text{Revision suggested,} & \text{defects are well supported.} \end{cases}$$

The exact thresholds should be learned and benchmarked empirically rather than asserted in advance.

22 The AI Review Certificate

A public label could help users understand the review state, but the wording must avoid confusing AI evaluation with peer review.

A suggested record is:

AI-Evaluated Research Object 6 heterogeneous evaluators Correctness assessment: high confidence Novelty assessment: probable Critical objections: 1 unresolved Verification status: partial Model disagreement: moderate Human expert review: pending

Another paper might say:

AI Review: Inconclusive The system could not reliably evaluate the central claim. Human expert review recommended.

The certificate is metadata about the *evaluation process*, not a statement of truth.

23 Empirical Evaluation Programme

A proposal of this magnitude should be tested against historical and prospective data.

23.1 Benchmark construction

Create a benchmark containing research papers with known outcomes and high-quality human assessments. The dataset should span multiple disciplines and include ordinary, borderline, rejected, accepted, highly cited, low-cited, interdisciplinary, and unconventional papers.

Critically, the benchmark should contain papers whose authors and institutions can be counterfactually reassigned.

23.2 Primary metrics

Valuable-work recall. Among papers later judged scientifically worthwhile by experts, what fraction survive AI triage?

$$\text{Recall} = \frac{TP}{TP + FN}.$$

False-negative rate.

$$FNR = \frac{FN}{TP + FN}.$$

This should be treated as a high-priority safety metric.

False-positive rate.

$$FPR = \frac{FP}{FP + TN}.$$

Calibration. If the system reports 80% confidence, future outcomes should approximate that level of reliability.

Expert-hours saved. Measure total human review time with and without AI triage.

Time-to-expert-attention. Measure how long it takes for a promising manuscript to receive the attention of a qualified expert.

Bias sensitivity. Measure

$$B_{\text{affiliation}}, B_{\text{seniority}}, B_{\text{gender}}, B_{\text{publication history}},$$

using blinded counterfactual experiments.

Disagreement quality. Evaluate whether AI disagreement predicts cases in which human experts themselves disagree.

Verification precision. For machine-generated objections, measure the fraction confirmed by independent experts or formal tools.

23.3 The most important benchmark: outsider recall

A special benchmark should contain manuscripts deliberately sampled from outside prestigious institutional networks or with prestige metadata removed. The goal is not to privilege outsiders but to test whether the system preserves their access to human scrutiny.

A useful hypothesis is:

$$\mathbb{P}(\text{expert escalation} \mid \text{scientific quality})$$

should be approximately invariant to irrelevant institutional metadata.

24 A Prospective Pilot Study

The proposal can be piloted without changing an entire journal.

24.1 Pilot design

A venue could take a historical collection of several hundred manuscripts and run the AI panel retrospectively. A second collection could be reviewed prospectively alongside ordinary review. Humans remain the final decision-makers.

The experiment would compare:

Human only versus Human + AI triage.

Primary outcomes would include review time, reviewer satisfaction, identification of technical flaws, missed valuable papers, and prestige-bias measures.

24.2 A stronger factorial design

An even stronger experiment would randomize manuscripts across:

blind vs. metadata-visible

and

human-only vs. AI-assisted.

This permits direct estimation of whether AI changes the effect of prestige cues.

25 Interdisciplinary Routing

A single general reviewer is particularly problematic for interdisciplinary work. We therefore propose a routing graph.

Let a manuscript have topical vector

$$\tau(P) \in \mathbb{R}^d.$$

A routing module maps this to a set of domain experts:

$$\rho(P) = \{e_1, e_2, \dots, e_m\}.$$

For a hybrid manuscript, the system may require multiple perspectives rather than one generalized reviewer. For example,

mathematics + control theory + signal processing

may warrant three specialized evaluations.

AI is well suited to performing the first-pass decomposition because it can read across vocabularies. Human experts are then assigned to verify the technically difficult parts.

26 The Economics of Attention

The economic argument for AI Review should be framed carefully. The point is not that all peer review costs exactly a particular dollar amount, nor that AI is free. The point is that machine inference is potentially much more scalable than human expert labor.

If one human review costs h hours and an AI pre-review costs c dollars, then a simple screening architecture is attractive when

$$\frac{c}{\text{expected human-hours saved}} \ll \text{marginal cost of expert labor.}$$

The exact ratio must be empirically measured in each deployment.

AAAI-26 provides one important real-world data point: its AI-review pilot processed 22,977 full-review papers in less than a day [1]. The fact that this was possible at all demonstrates that the scale mismatch between machine inference and human review is operationally significant.

The correct economic objective is therefore not “replace reviewers with AI.” It is:

reduce routine human workload while preserving high-value human judgment.

27 The Relationship to Open Science

The proposed architecture is naturally compatible with open science because it favors persistent, inspectable evidence.

A paper’s scientific state becomes a graph rather than a single PDF:

$$\mathcal{G}(P) = (V, E),$$

where vertices represent manuscript versions, reviews, objections, data sets, code, replications, and certifications; edges represent relations such as “criticizes,” “responds to,” “verifies,” “extends,” or “supersedes.”

Such a graph can capture scientific development more faithfully than a single publication event.

An especially important consequence is that negative information becomes useful. If a reviewer finds a flaw and the author repairs it, the final object retains the history:

$$\text{claim} \rightarrow \text{objection} \rightarrow \text{repair} \rightarrow \text{verification}.$$

This can be more informative than a binary “accepted” label.

28 Reviewer Reputation and Expert Attention

A decentralized system needs incentives for expert participation. However, popularity must not replace expertise.

We propose that the platform maintain reviewer records based on evidence quality, not merely review count.

Possible signals include:

- proportion of criticisms later confirmed;
- proportion of positive claims later supported by independent evidence;
- specificity of technical comments;
- domain relevance;
- conflict-of-interest history;
- author responses to the review;
- other expert assessments of the reviewer’s reasoning.

One should not construct a single global “reviewer score” without qualification. Expertise is local. A brilliant algebraist is not automatically an authority on a clinical trial.

The desirable structure is therefore a domain-specific reputation vector:

$$\text{Rep}(e) = (r_1, r_2, \dots, r_d).$$

29 Governance Principles

The following principles are proposed as a minimum constitutional framework for AI Review.

1. **Human non-replacement:** AI Review does not by itself certify scientific truth.
2. **Blind first pass:** irrelevant author-status metadata should be withheld initially.
3. **Heterogeneity:** no single model should control the evaluation pipeline.
4. **Evidence requirement:** consequential criticisms should point to checkable evidence.
5. **Abstention:** uncertainty must be representable.
6. **Human appeal:** a machine classification cannot be an irreversible scientific veto.
7. **Transparency:** model identity, rubric, date, tools, and substantive outputs should be archived where confidentiality permits.
8. **Security:** manuscripts are untrusted input and must be sandboxed.
9. **Privacy:** confidential manuscripts require controlled processing and explicit governance.
10. **Versioning:** corrections should create new states rather than erase prior evidence.
11. **Auditability:** deployed systems must publish performance and bias measurements.
12. **Modularity:** journals, societies, preprint servers, and communities should be able to adopt or reject the layer independently.

30 What the Proposal Explicitly Rejects

To avoid misunderstanding, this paper does *not* propose the following.

30.1 It does not propose “AI decides truth”

No present AI system should be treated as a general scientific oracle.

30.2 It does not propose abolishing peer review

Human review remains necessary, especially for consequential or technically specialized claims.

30.3 It does not propose replacing experts with majority vote

Five agreeing models may share an error. Likewise, ten inexperienced human readers do not necessarily outweigh one expert’s verified counterexample.

30.4 It does not propose treating independent researchers as inherently superior

The goal is content-based scrutiny, not anti-academic favoritism.

30.5 It does not propose eliminating journals immediately

Journals can remain valuable certification, editorial, and archival institutions.

30.6 It does not claim current AI systems are sufficiently mature

The proposal is deliberately designed to make that question empirical.

31 A Concrete End-to-End Protocol

A minimal operational protocol can be written as follows.

1. **Submission:** the author submits a versioned research object.
2. **Sanitization:** author prestige metadata is separated from the technical manuscript.
3. **Routing:** the system identifies relevant scientific domains and verification tools.
4. **Independent AI passes:** heterogeneous evaluators independently review the manuscript.
5. **Adversarial pass:** at least one evaluator is assigned to search for fatal defects.
6. **Evidence retrieval:** citations and relevant prior work are checked.
7. **Tool verification:** symbolic, numerical, statistical, or code-based checks are attempted.
8. **Cross-examination:** evaluators challenge one another's criticisms.
9. **Adjudication:** a separate agent or small panel summarizes unresolved disagreements.
10. **Triage:** the manuscript receives a transparent status: promising, problematic, or inconclusive.
11. **Human routing:** promising and inconclusive papers may be routed to appropriate experts; problematic papers can be revised and resubmitted.
12. **Expert review:** human reviewers assess specific claims and objections.
13. **Author response:** the author may respond to critiques and revise the manuscript.
14. **Evidence update:** the critique ledger is updated rather than erased.
15. **Optional certification:** the author may submit to a journal, society, conference, or other certifying institution.

32 Pseudocode for a Reference Implementation

The following pseudocode illustrates the architecture without committing it to a particular vendor or model.

```
function AI_REVIEW(paper):
    clean_paper = strip_irrelevant_identity_metadata(paper)
    domain = infer_domains(clean_paper)
    tools = route_verification_tools(domain)
```

```
reviews = []
for evaluator in heterogeneous_panel:
    reviews.append(
        independent_review(
            clean_paper,
            role=evaluator.role,
            tools=tools,
            blind=True
        )
    )

objections = extract_falsifiable_objections(reviews)
evidence = verify_objections(objections, tools)

debates = cross_examine(
    reviews=reviews,
    objections=objections,
    evidence=evidence
)

adjudication = adjudicate(debates, evidence)
profile = aggregate_with_disagreement(
    reviews,
    adjudication
)

if verified_fatal_issue(profile):
    status = "PROBLEMATIC"
elif high_uncertainty(profile) or high_disagreement(profile):
    status = "INCONCLUSIVE"
else:
    status = "PROMISING"

return EvaluationRecord(
    status=status,
    profile=profile,
    objections=objections,
    evidence=evidence,
    debates=debates,
    adjudication=adjudication,
    model_versions=versions,
    rubric_version=rubric.version
)
```

33 A Mathematical View of False-Negative Risk

The strongest argument for a triage system rather than a hard gate can be formalized.

Let $G(P) = 1$ indicate that a sufficiently broad expert panel ultimately considers manuscript P worthy of serious scientific attention, and let $A(P) = 1$ indicate that AI triage escalates it to humans.

The harmful false-negative probability is

$$\text{FN} = \mathbb{P}(A(P) = 0 \mid G(P) = 1).$$

Suppose the system is intended to protect unconventional research. We can add a subgroup indicator $O(P) = 1$ for manuscripts outside conventional institutional prestige structures. A fairness-oriented safety analysis should examine

$$\text{FN}_O = \mathbb{P}(A(P) = 0 \mid G(P) = 1, O(P) = 1)$$

and compare it with

$$\text{FN}_{\neg O} = \mathbb{P}(A(P) = 0 \mid G(P) = 1, O(P) = 0).$$

The target should not be that the outsider subgroup receives preferential treatment. The target should be that scientifically valuable work does not face a large additional false-negative burden merely because of irrelevant status.

Ideally,

$$\text{FN}_O \approx \text{FN}_{\neg O}.$$

This formulation converts an ethical intuition into a measurable property.

34 A Bayesian Interpretation

Another useful framing is sequential evidence accumulation.

Let H denote the hypothesis that a paper deserves serious expert attention. Before evaluation, the system has prior odds

$$\frac{\mathbb{P}(H)}{1 - \mathbb{P}(H)}.$$

Each independent-ish piece of evidence E_i updates the odds:

$$\frac{\mathbb{P}(H \mid E_1, \dots, E_n)}{1 - \mathbb{P}(H \mid E_1, \dots, E_n)} = \frac{\mathbb{P}(H)}{1 - \mathbb{P}(H)} \prod_i \frac{\mathbb{P}(E_i \mid H)}{\mathbb{P}(E_i \mid \neg H)}.$$

In reality the evidence sources are dependent, so the naïve product must not be taken literally. Nevertheless, the conceptual lesson is useful: an AI review is one evidence source among many.

A verified proof, a successful replication, and a detailed expert confirmation should contribute different evidentiary weights from six stylistically similar LLM outputs.

This motivates the notion of an *evidence hierarchy* rather than a vote count.

35 Evidence Hierarchy

A practical hierarchy might be:

Level	Example
0	Unchecked model impression
1	Multiple AI evaluations agree
2	AI criticism linked to explicit evidence
3	Independent computational or symbolic verification
4	Substantive expert verification
5	Independent human reproduction or extension
6	Broad subsequent scholarly confirmation

The numbers are not meant to be a universal scientific scale. They illustrate a principle: *independence and verification should matter more than rhetorical confidence.*

36 The Case for Open Expert Commentary

Traditional double-blind review can protect reviewers and authors, but it can also make the intellectual path to acceptance opaque. An open evidence system provides a different trade-off.

A researcher may choose to publish the full expert comments, author responses, and verification history. Another venue may choose to keep some expert identities private. The architecture should support both.

A valuable compromise is “open argument, controlled identity”: the technical reasoning is public, while personal information is disclosed according to governance rules.

This would allow a reader to see:

Reviewer objection → author response → adjudication → current status.

The result is not a popularity contest but a public technical record.

37 Why Journals Could Still Benefit

At first sight a decentralized evidence layer may appear threatening to publishers. It need not be.

A journal can treat the evidence graph as input. Instead of spending editorial time discovering obvious errors, it may receive:

- structured AI critiques;
- verified computational outputs;
- literature comparisons;
- documented disagreements;
- author responses.

Human editors can then spend more time on substantive judgments such as novelty, significance, community fit, and final certification.

In this sense, AI Review could be a cost-saving technology for journals without being a threat to their existence.

The decentralization claim is therefore not “destroy publishers.” It is:

do not require one institution to monopolize the entire evaluation chain.

38 Relationship to Existing AI-Review Research

The proposal is deliberately positioned within, rather than outside, the rapidly developing AI-for-peer-review literature.

The 2026 survey by Wu et al. maps a broad landscape including fine-tuned reviewers, agent-based systems, reinforcement-learning approaches, retrieval augmentation, rebuttal generation, meta-review synthesis, and revision loops [5]. This paper differs in emphasis. Rather than primarily asking how to generate a better review text, it asks how to design an *institutional evaluation layer* that connects open dissemination, AI triage, expert attention, and optional journal certification.

AAAI-26 provides evidence that multi-stage AI review can operate at conference scale [1]. The present proposal treats large-scale deployment as evidence for feasibility, not as proof of full scientific reliability.

Douven’s 2026 heterogeneous-panel study provides empirical support for panel diversity and comparative selection [2]. The present architecture extends that idea with explicit abstention, adversarial verification, bias audits, and a human appeal channel.

Vasu et al. demonstrate that LLM review can exhibit affiliation and seniority biases [3]. This paper takes that as a reason to make blind review and counterfactual auditing constitutional requirements.

The ACL position paper on responsible AI review recommends moving toward meaningful tasks such as reproducibility and citation verification [6]. The present proposal adopts that recommendation as a core rather than peripheral component.

The central research opportunity therefore lies not in proving that “AI is better than humans,” but in combining the strongest emerging components into an auditable workflow.

39 Possible Objections and Responses

39.1 Objection: LLMs do not understand science

This is a serious objection if the system is treated as an autonomous scientific authority. It is less decisive if the system is treated as a triage and verification orchestrator. Code can execute. Algebra can be checked. Citations can be retrieved. Contradictory arguments can be surfaced. The system need not possess complete scientific understanding to perform useful sub-tasks.

39.2 Objection: AI will reproduce the biases of academia

Likely true unless explicitly tested. That is why the proposal requires blind evaluation, counterfactual audits, heterogeneous models, and public reporting of bias metrics. The goal is not to assume neutrality but to make bias measurable.

39.3 Objection: authors will game the system

Likely true. Any published rubric can become a target. This is another reason not to reduce evaluation to one score. The system should evolve its prompts, verify claims, run adversarial tests, and keep some evaluation diversity hidden until after submission.

39.4 Objection: AI review adds another gatekeeper

It can, if poorly designed. The right to human appeal, the inconclusive state, and the rule that AI may route but not irreversibly veto are therefore structural safeguards.

39.5 Objection: only experts can recognize genius

Sometimes true. The architecture explicitly preserves expert review. The purpose of AI is to reduce the number of manuscripts requiring initial human attention, not to eliminate expert access.

39.6 Objection: peer review contains valuable social knowledge that AI cannot reproduce

Correct. Informal networks, disciplinary memory, experimental craft, and tacit knowledge are important. The system should therefore be seen as augmentation. A human expert's domain judgment remains indispensable in cases where explicit text and available tools do not capture the relevant context.

39.7 Objection: decentralized publication would create chaos

Possibly, if all evidence were treated equally. The proposal instead introduces explicit states, evidence levels, reviewer provenance, versioning, and certification options. Decentralization need not mean the absence of standards; it can mean multiple transparent standards rather than one opaque gateway.

40 A Proposed Minimum Standard for AI Review

A platform should not advertise a manuscript as "AI reviewed" unless it satisfies minimum requirements.

1. At least three heterogeneous evaluation systems or demonstrably independent evaluation pathways.
2. A published rubric appropriate to the research domain.
3. Blind-first evaluation with respect to irrelevant author prestige.
4. Explicit model disagreement reporting.
5. At least one adversarial evaluation pass.
6. Evidence-linked criticisms.
7. Tool-assisted verification where technically possible.
8. An abstention or inconclusive category.
9. Human appeal or escalation.
10. Versioned storage of evaluation records.
11. Public disclosure of model versions and evaluation date, subject to security and confidentiality constraints.
12. Periodic independent bias and calibration audits.

A system failing these conditions may still be useful, but it should not claim equivalence to the proposed protocol.

41 A Maturity Model

AI Review could evolve through maturity levels.

Level	Capability
L0	Single-model prose generation
L1	Multi-model independent review
L2	Structured disagreement and adversarial review
L3	Retrieval and tool-assisted verification
L4	Blind counterfactual bias auditing
L5	Expert routing and persistent critique ledger
L6	Decentralized evidence graph and cross-platform certification

The proposed ecosystem should not wait for Level 6 before experimentation. Early pilots can test Levels 1–3 while explicitly refusing to overclaim.

42 Limitations of the Present Proposal

The proposal itself has limitations.

First, its strongest claims are architectural and normative rather than experimentally demonstrated. The empirical results cited from 2026 show that AI review is feasible at scale and that heterogeneous evaluation can be useful, but they do not prove that the exact architecture proposed here improves scientific discovery across all disciplines.

Second, the paper is necessarily more mature for text-dominant fields than for fields requiring physical experiments, specialized laboratory equipment, fieldwork, or tacit procedural knowledge.

Third, “independent” models may not be truly independent. Shared training data and common scientific conventions can create correlated errors.

Fourth, open review can create retaliation, social pressure, and reputational risks. Decentralization does not automatically produce fairness.

Fifth, access to advanced AI systems may itself become unequal. If only wealthy organizations can run large review panels, the system could reproduce the infrastructure inequality it was intended to reduce. Open-weight alternatives and shared public infrastructure will therefore be important.

Sixth, the proposed evidence hierarchy is conceptual. Empirical calibration is required before weights or thresholds are assigned.

43 Research Agenda

A serious research program around this proposal should proceed in stages.

43.1 Stage I: Historical validation

Build benchmark corpora with known human outcomes and expert assessments.

43.2 Stage II: Bias auditing

Perform counterfactual affiliation, seniority, gender, nationality, and publication-history interventions.

43.3 Stage III: Verification benchmarking

Measure whether AI-generated mathematical, statistical, and computational objections are correct.

43.4 Stage IV: Human workload study

Measure whether AI triage actually reduces reviewer time without lowering valuable-work recall.

43.5 Stage V: Prospective deployment

Run AI Review alongside ordinary peer review at a willing conference, journal, repository, or research society.

43.6 Stage VI: Open infrastructure

Release anonymized evaluation records, prompts where safe, model versions, calibration reports, and benchmark results.

43.7 Stage VII: Longitudinal scientific validation

Track whether papers flagged as promising actually generate expert confirmation, replication, extensions, or sustained scholarly use.

The final criterion should be downstream scientific performance, not aesthetic quality of AI-generated reviews.

44 A Possible Institutional Future

The most plausible future is not one in which journals disappear overnight. It is one in which scientific legitimacy becomes multi-layered.

A research object may have:

Dissemination status:	Open preprint
AI status:	AI evaluated
Expert status:	Independently reviewed
Replication status:	Reproduced
Certification status:	Journal certified.

These labels answer different questions. A reader can choose which evidence level matters for their purpose.

This model may be particularly valuable for mathematics, theory, computer science, and interdisciplinary work where a manuscript can often be inspected using symbolic, computational, or textual tools before scarce human attention is spent.

It may also permit a new social pattern. A researcher outside the dominant institutional network could release a manuscript. A machine panel could identify a nontrivial theorem. An expert could discover the result through the evidence graph. The expert could verify one lemma and challenge another. The author could respond. The research object could accumulate credibility before ever entering a conventional journal.

This does not guarantee discovery of overlooked genius. No system can. It merely reduces the number of unnecessary gates between a claim and potential scrutiny.

45 A Note on the Meaning of “Good Enough”

The phrase “good enough alternative” is deliberately provocative but should be interpreted carefully.

A machine need not achieve perfect agreement with experts to be useful. Suppose AI Review raises the probability that a valuable paper receives expert attention from p_0 to p_1 , with

$$p_1 > p_0,$$

while reducing routine human review burden. Then it has created value even if it still makes errors.

Likewise, a system can be useful if it catches obvious defects before submission. An author who learns that a theorem has a missing hypothesis, that a citation is wrong, or that a numerical result does not reproduce has received a scientifically useful service even if no journal ever sees the manuscript.

This is why the standard for success should be comparative rather than absolute:

AI-assisted process versus existing process.

The right question is not:

Is AI perfect?

It is:

Does a carefully governed AI layer improve the scientific evaluation ecosystem enough to justify its residual risks?

That is an empirical question.

46 Conclusion

Traditional peer review remains one of the central institutions of modern science. Its problems do not imply that it should be abolished, just as the limitations of current AI do not imply that machines should be trusted as scientific oracles. The more productive possibility lies between those extremes.

This paper proposes such a middle layer: AI-Assisted Open Peer Review.

The proposed system begins with open dissemination rather than institutional permission. It performs a blind first-pass evaluation using heterogeneous AI systems. It separates correctness, novelty, significance, rigor, evidence, and reproducibility rather than compressing everything into one score. It assigns specialized roles to evaluators, introduces an adversarial referee, retrieves and verifies literature, executes computational or symbolic checks when possible, exposes disagreement instead of hiding it, and records a persistent critique ledger. It then routes promising or inconclusive work toward human experts while retaining a right to human appeal. Finally, it permits journal publication to remain as a valuable certification layer without making the journal the only place where meaningful evaluation can occur.

The proposal has a simple constitutional principle:

**AI should help allocate human scientific attention, not
monopolize it.**

This principle has several consequences. A model should not possess an irreversible veto. A numerical AI score should not become a new prestige credential. Model agreement should not be confused with truth. Author identity should not contaminate the initial content evaluation. Hostile comments should be judged for technical substance separately from tone or status dependence. And every machine judgment should be understood as evidence whose reliability must itself be measured.

The proposal is deliberately compatible with existing institutions. It does not require journals to disappear. It does not require human peer review to disappear. It requires only that the scientific community permit additional routes by which a research object can acquire scrutiny, evidence, and visibility.

The deeper objective is therefore not to replace peer review but to redistribute the burden of evaluation. Human expertise is scarce. AI inference is increasingly scalable. Open repositories can disseminate work at low marginal cost. Verification tools can check some classes of claims automatically. Expert attention can then be concentrated where it is most needed.

If implemented carefully, AI Review could become neither an automated acceptance machine nor an adversarial substitute for scientists. It could instead become something more modest and perhaps more useful: an always-available first line of scrutiny, a detector of obvious failures, a challenger of weak arguments, a verifier of computational claims, a retrieval engine for prior work, an organizer of disagreement, and a routing mechanism that helps promising research reach humans who can genuinely judge it.

The decisive test, however, is not rhetorical. It is empirical. The proposed ecosystem should be measured by whether it reduces human workload without sacrificing scientific quality, whether it lowers false-negative rates for unconventional work, whether its evaluations remain robust under counterfactual prestige changes, whether its criticisms are actually correct, and whether the resulting evidence record improves the discoverability and verification of research.

Science should not become a contest between humans and machines. Nor should it become a contest between journals and preprints. A better goal is a system in which dissemination, critique, verification, expertise, and certification can coexist as distinct but connected layers.

In that system, the question would no longer be only:

Has a journal accepted this paper?

It would also be:

What evidence has this research accumulated, who has examined it, what objections have survived, what has been independently verified, and how confidently should the scientific community investigate it further?

That is the question an open, AI-assisted, human-governed research ecosystem should be built to answer.

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Author's Note

This preprint is motivated in part by the practical observation that access to scientific evaluation is not distributed uniformly. An unaffiliated researcher may be able to formulate, compute, and disseminate ideas while lacking the institutional resources that normally surround formal publication. The proposal developed here is therefore intentionally designed to distinguish the scientific quality of a research object from the institutional status of its creator.

The author is Pearl Bipin Pulickal, a graduate engineer in Electronics and Communication Engineering from the National Institute of Technology Goa and an independent researcher. The author does not claim that independence is an epistemic virtue in itself. The claim is narrower: scientific communities benefit when research can be scrutinized on its technical merits even when its creator does not possess conventional institutional status.

The present work should therefore be read as a proposal for infrastructure rather than as a rejection of scholarship, expertise, or peer review. Its central aspiration is to create more paths from an idea to serious scrutiny.